

XIV FORUM OF PARTIAL DIFFERENTIAL EQUATIONS

Book of Abstracts

Test edition – four contributions

14–18 September 2026

Będlewo Conference Center, Poland

<https://xivfpde.com>

About this test edition

This test edition demonstrates the proposed typographic structure for the final *Book of Abstracts* of the XIV Forum of Partial Differential Equations. The wording of the abstracts is reproduced from the authors' submitted PDF files; only typographic markup and page layout have been standardized.

The sample deliberately combines contributions with different levels of mathematical notation and bibliography complexity. Each abstract begins on a new page so that the layout can be reviewed independently before the complete collection is prepared.

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INVITED LECTURE

Defying Scaling in Potential Theory

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Abstract

Classical non-linear potential theory thrives under the predictable comfort of power-law scaling, where p -Laplacian operators yield well-behaved pointwise estimates through traditional Wolff potentials. But what happens when homogeneity breaks down completely? In this talk, we defy the constraints of standard scaling to tackle measure data elliptic problems governed by second-order operators with non-standard Orlicz growth and merely measurable coefficients.

We break away from homogeneous paradigms by establishing sharp pointwise estimates expressed in terms of novel, generalized potentials tailored to non-scaling structures. Going beyond describing local solution behavior where standard tools fail, we uncover the exact regularity consequences dictated by these estimates. Crucially, this framework holds across both scalar equations and vectorial systems, forging a powerful potential theory liberated from scaling laws.

CONTRIBUTED TALK

Methods of the Conley index for a class of functionals in Banach spaces involving p -Laplacian

Marek Izydorek

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Abstract

We will be concerned with a class of functionals involving the p -Laplacian. The natural setting for these functionals is the Banach space $X = W_0^{1,2p}(\Omega)$, where $p > n/2$ and $\Omega \subset \mathbb{R}^n$ is a bounded domain with sufficiently smooth boundary.

To a given functional $f : X \rightarrow \mathbb{R}$, we associate a quasilinear elliptic problem (EL), defined by the corresponding Euler-Lagrange equation. The critical points of f are precisely the solutions of (EL). Since X is not isomorphic to its dual space, the standard notion of nondegeneracy of critical points is not applicable in this setting. Moreover, f serves as a Lyapunov function for a vector field which, in our setting, is merely continuous and therefore need not generate a flow.

We develop a method based on an extension of the Conley index to Banach spaces for merely continuous vector fields. This framework allows us to overcome the difficulties arising from the lack of a flow and the absence of the usual nondegeneracy condition, and to establish existence and multiplicity results for solutions of (EL) in the presence of $\mathbb{Z}_2$ -symmetry.

CONTRIBUTED TALK

A method to determine the minimal null control time of one-dimensional linear hyperbolic systems

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Abstract

How fast can a one-dimensional transport process be brought to rest by acting on just one end? This is a question about the *minimal null control time*, a central notion in control theory that, for coupled hyperbolic systems, turns out to be delicate.

Here, we consider first-order linear hyperbolic systems of the form:

$$\begin{cases} \frac{\partial y}{\partial t}(t, x) + \Lambda(x) \frac{\partial y}{\partial x}(t, x) = M(x)y(t, x), \\ y_-(t, 1) = u(t), & y_+(t, 0) = Qy_-(t, 0), \\ y(0, x) = y^0(x), \end{cases} \quad (t, x) \in (0, T) \times (0, 1).$$

The components propagate at the speeds prescribed by Λ , interact through the internal coupling M , are coupled at the boundary through Q , and are steered by a control u acting on one side. Such systems arise as linearizations, around steady states, of the nonlinear balance laws governing many transport phenomena [BC16; YK22]. Already in this linear setting, determining the minimal control time is a subtle problem, as it depends not only on the propagation speeds but on the couplings.

In this talk, we present a general method that computes this minimal control time exactly, whenever Λ and M are sufficiently smooth functions of x . This presentation is based on the joint work [HO25] with Long Hu.

References

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CONTRIBUTED TALK

Higher regularity for relaxed micromorphic models: from smooth to Lipschitz domains

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Abstract

The relaxed micromorphic model is a generalized continuum model designed to describe the effective behaviour of microstructured materials and metamaterials. It couples the classical displacement field with a non-symmetric microdistortion tensor P and, from the analytical point of view, combines an elasticity-type system with a Maxwell-type system.

A characteristic feature of the model is that the curvature energy depends on $\text{Curl } P$ rather than on the full gradient ∇P . Consequently, weak solutions are naturally sought in the space

$$H(\text{Curl}; \Omega) = \left\{ P \in L^2(\Omega) : \text{Curl } P \in L^2(\Omega) \right\},$$

while the energy does not directly control all weak derivatives of P . This naturally raises the question of whether one can recover control of the full gradient ∇P from the structure of the system.

I will first briefly introduce the relaxed micromorphic model and recall earlier existence and higher-regularity results, including local regularity and global H^1 -regularity of the microdistortion on smooth domains. The main part of the talk will be devoted to a global regularity result for a physically nonlinear version of the model on bounded Lipschitz domains. Since standard difference-quotient arguments are not suitable in this setting, the proof is based on modified finite differences constructed from inner variations together with a Piola-type transformation adapted to the Curl structure. This yields, for every $\varepsilon > 0$,

$$u \in H^{\frac{3}{2}-\varepsilon}(\Omega), \quad P, \text{Curl } P \in H^{\frac{1}{2}-\varepsilon}(\Omega).$$

The results presented in the talk are based on joint work with Dorothee Knees and Patrizio Neff.

References

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